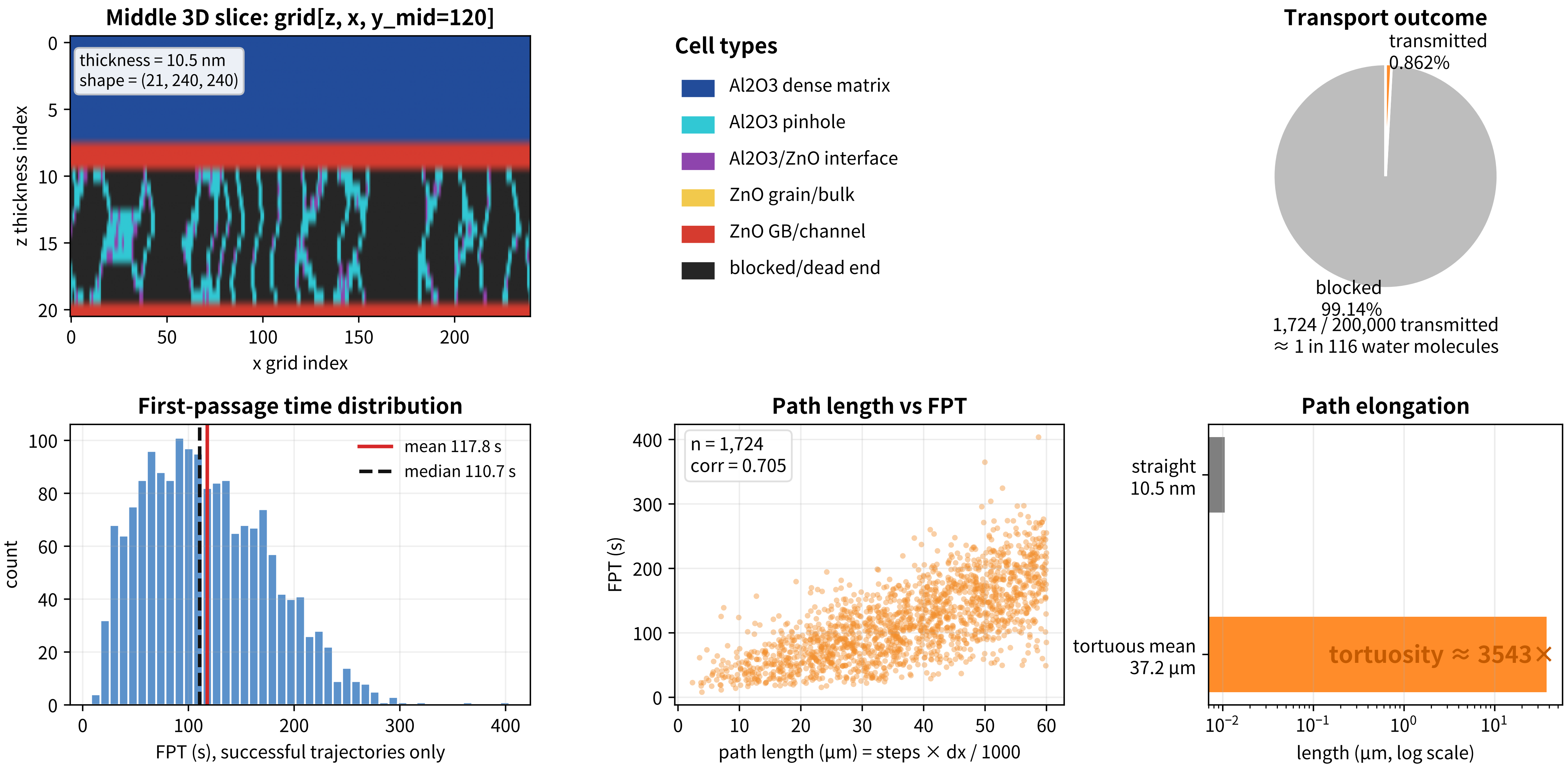
3D Al2O3/ZnO bilayer kMC: readable transport analysis

Rare Al2O3 pinholes + interface lateral search + ZnO boundary/channel tortuosity make successful transmission uncommon.



P_trans	mean FPT	FPT p10–p90	mean path	path p10–p90	pinhole area
0.862% (1,724/200,000)	**117.8 s**	**44.2–201.2 s**	**37.2 µm**	**17.2–55.8 µm**	**0.0677%**

Mechanism: Most water molecules do not enter because Al2O3 pinholes are rare. A successful molecule must find a pinhole, search laterally at the Al2O3/ZnO interface, then follow a long boundary/channel path while avoiding blocked dead ends. Therefore geometric thickness is only 10.5 nm, but successful paths average ~37.2 µm.